Report on Project

Smart Fault Detection Glasses (DivyaDrishti)

Abstract

DivyaDrishti is an AI-powered smart inspection system designed to detect defects in industrial components in real time. It leverages computer vision, edge computing, and cloud integration to reduce human error and improve efficiency in manufacturing environments.

Problem Statement

Manual inspection in industries is repetitive, time-consuming, and prone to human error due to fatigue and repetitive tasks. This often leads to:

- Missed defects
- Inconsistent inspection results
- Increased operational costs
- Reduced product quality

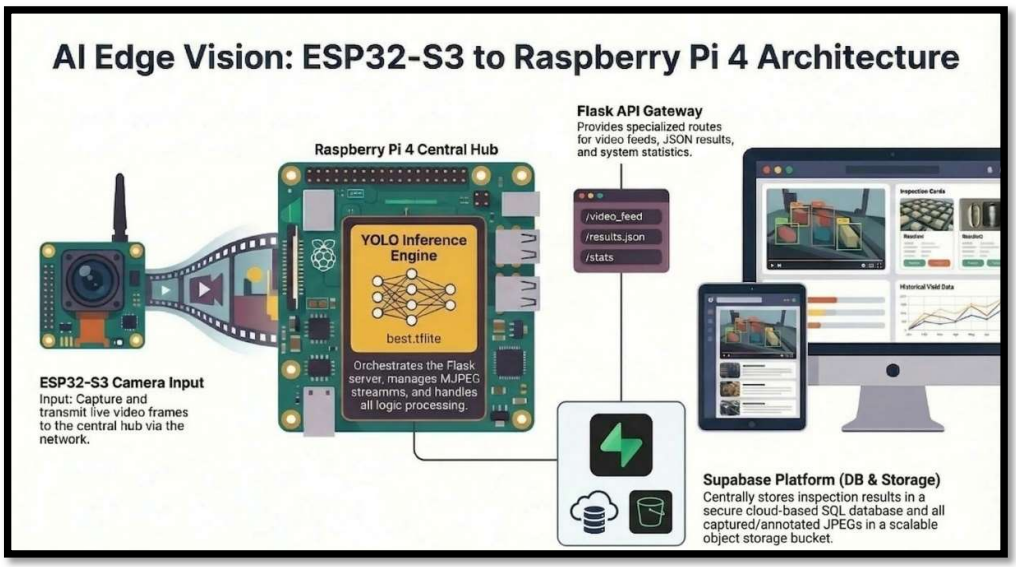
Proposed Solution

We developed DivyaDrishti – Smart Fault Detection Glasses, an AI-based system that performs real-time inspection using a camera and machine learning.

The system:

- Captures images using a camera
- Processes them using an AI model
- Classifies results as **OK / NOK**
- Sends data to cloud
- Displays results on a live dashboard

System Architecture



Hardware Components

- Raspberry Pi 4 Model B
- Camera Module / Webcam

- Adapter

Software Technologies

- AI & Processing
  - Python
  - OpenCV
  - TensorFlow / CNN
- Frontend
  - React.js
  - Firebase SDK
- Backend / Cloud
  - Firebase Firestore (Realtime database)
  - Supabase Storage (Image hosting)

Flow Diagram

1. Camera captures image of component
2. Image is processed by AI model
3. Defect is detected and classified (OK/NOK)
4. Image uploaded to Supabase
5. Metadata stored in Firebase Firestore
6. Dashboard updates in realtime
7. Alert triggered if defect detected

Features

- Realtime defect detection
- Edge AI processing (low latency)
- Live dashboard monitoring
- Image storage and retrieval
- Alert system (SMS / WhatsApp)
- Scalable cloud architecture

Cost Efficiency

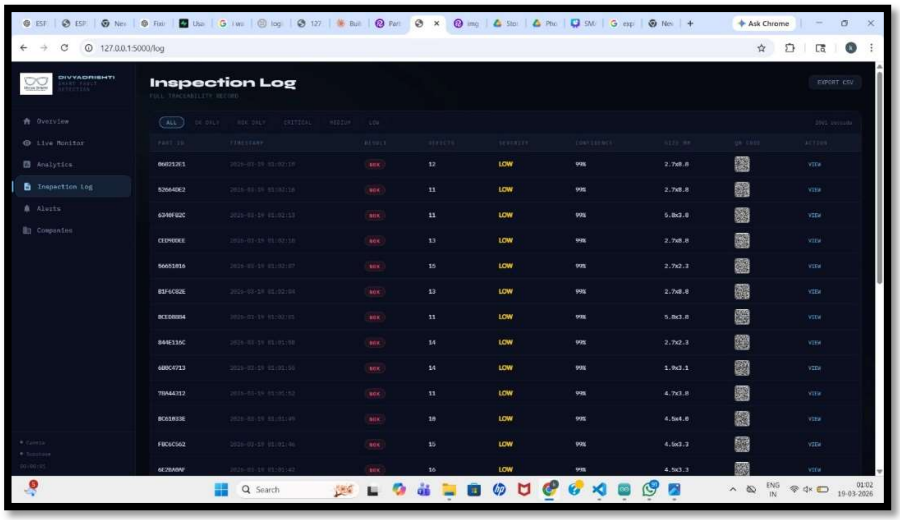
- Uses lowcost hardware (Raspberry Pi)
- Uses freetier cloud services
- Reduces dependency on expensive industrial systems

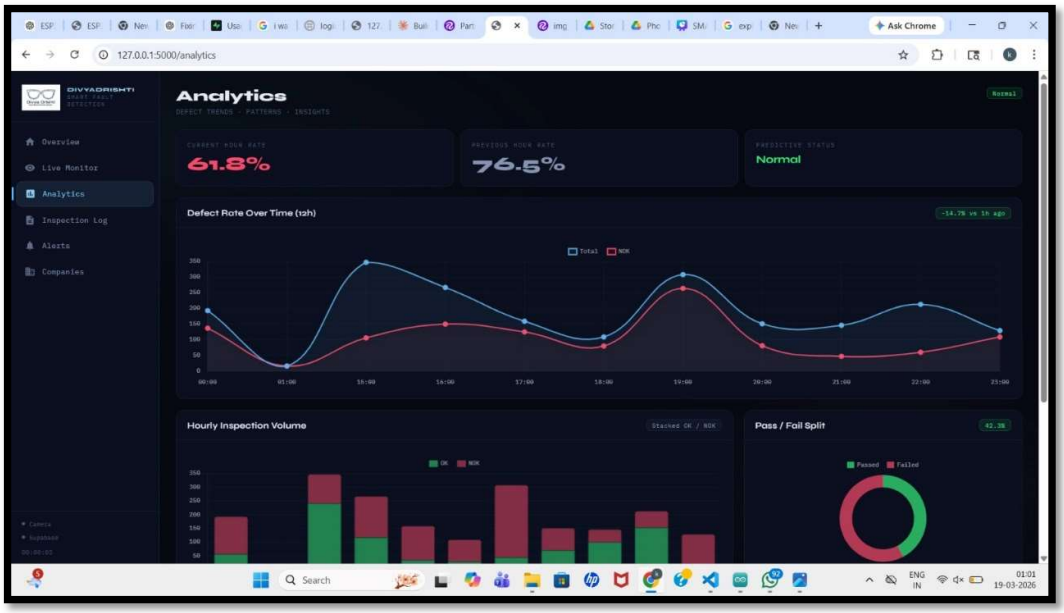
Results & Analysis

System performance evaluation includes:

- Detection accuracy
- Processing speed

- Response time





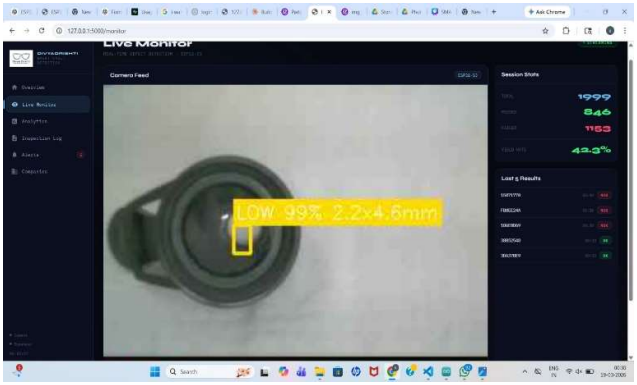
Examples:

Accuracy vs Dataset size

Defect detection rate

Processing latency

Screenshots



Include:

- Dashboard UI
- Live detection output
- Alert system
- Image preview

Market Potential

This solution can be applied in:

- Manufacturing industries
- Automotive sector
- Electronics production
- Quality control pipelines

Business Impact

- Improved product quality
- Reduced defect leakage
- Faster inspection process
- Reduced labor dependency
- Datadriven decision making

Scalability

- Supports multiple devices (horizontal scaling)
- Cloud backend autoscales
- Multifactory support
- Edge processing reduces load

Future Scope

- Integration with actual smart glasses
- Defect localization (highlighting defects)
- Predictive maintenance using AI
- Mobile application development
- Advanced analytics dashboard

Conclusion

DivyaDrishti demonstrates how AI + IoT + Cloud can revolutionize industrial inspection systems by making them:

- Faster
- More accurate
- Scalable
- Cost-effective

Team Information:

Team Name: DivyaDrishti
Team Members:

- Kavisha Sharma
- Rudra Patel
- Sharmishta Kola
- Preet Desai
- Rahul Pal